Contents

Cross-Closure Theorems for Relation-First Organization Structures

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Foundation Statement

This document is not a new foundational theory. It is a cross-closure theorem note. Its purpose is to show that the existing companion theories do more than share vocabulary: under declared bridge conditions, structures generated in one closure context induce structures in another closure context.

The inherited companion theories are:

Theory of Organization

Charted Organization Theory

Recursive Chart Learning Theory

Triadic Closure and Recursive Residual Theory.

The central claims are conditional. A triadic closure schema does not automatically generate a charted organization. A closure residual does not automatically generate learning pressure. A persistence interval does not automatically generate a valid learning trace. Each implication requires explicitly declared bridge data.

This conditionality is not a defect of the framework. It is the cross-closure form of the task-visible declaration discipline developed in the companion methodological papers. A structure can cross from one closure context into another only when the relevant bridge data are declared. The bridge data are therefore part of the theorem, not auxiliary prose.

The main results are:

1. **Schema-to-chart theorem.** A boundary–mediation–visibility schema satisfying chartability conditions induces a charted organization structure.
2. **Charted organization inheritance theorem.** Once a schema induces a chart, the proper-organization, persistence, and no-go machinery of the Theory of Organization applies chartwise.
3. **Residual-to-learning-pressure theorem.** A proper closure residual can induce learning pressure only when it is declared as a faithful task-loss readout over a chart-visible branch field.
4. **Persistence-compatible trace theorem.** A learning trace remains proper only when its threshold policy remains inside the transported persistence supports of the organizations it claims to preserve.
5. **Three-chart mediation theorem.** Direct and mediated chart transport agree exactly when the corresponding mediation residual is neutral; when they differ, same-endpoint traces need not carry the same organization.
6. **Operation–trace–closure theorem.** Closure-depth is generated by admissible operation traces and is naturally a preorder, not a primitive numerical index.
7. **Cross-closure failure theorem.** Failure of chartability, faithfulness, persistence-compatibility, mediation-coherence, or declared closure generation blocks the corresponding cross-closure interpretation.

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0. Scope

0.1 What this document does

This document establishes conditional bridges between existing closure contexts in the programme. It does not introduce new primitives. It defines bridge data and proves that, when those data are supplied, structures generated by one companion theory induce structures governed by another.

0.2 What this document does not do

This document does not claim:

1. every triadic structure is chartable;
2. every residual is a learning loss;
3. every learning trace is proper;
4. every three-chart comparison has a neutral mediation residual;
5. every charted organization has a physical realization;
6. every external triadic theory instantiates this programme;
7. Lorentzian signature, physical gauge groups, consciousness, language, or biological learning have been derived.

0.3 Bridge discipline

A cross-closure implication is valid only when the relevant bridge data are declared. In particular:

$$\text{triadic schema} \not\Rightarrow \text{charted organization}$$

without chartability data;

$$\text{closure residual} \not\Rightarrow \text{learning pressure}$$

without faithful task-loss readout data;

$$\text{chart transition} \not\Rightarrow \text{proper learning trace}$$

without persistence-compatible threshold data.

0.4 Bridge declaration discipline

This note uses the following discipline throughout. A cross-closure inference from a source context X to a target context Y is not licensed by vocabulary alignment alone. It requires declared **bridge data**:

$$\mathfrak{B}_{X\to Y}.$$

The expression

$$X + \mathfrak{B}_{X\to Y} \Rightarrow Y$$

means that a structure of type X, together with adequate bridge data of the declared kind, induces the target structure Y. The bridge data may include chartability maps, faithful residual readouts, persistence-compatible threshold policies, organization transports, or operation systems, depending on the theorem.

This is the same kind of declaration discipline as the task-visible quotient criterion: preservation does not cross a quotient, chart, residual, or trace boundary unless the preservation data have been declared. The difference is only the scale. The quotient criterion governs preservation inside a single task setting; bridge declaration governs preservation across closure contexts.

0.5 Cross-closure theorem clusters

The term **cross-closure** refers to the full set of bridge theorem clusters in this note, not only to the operation–trace–closure construction in Section 12. The note contains several distinct cross-closure moves:

1. a schema-to-chart bridge;
2. a chart-to-organization inheritance bridge;
3. a residual-to-learning-pressure bridge;
4. a persistence-compatible trace bridge;
5. a three-chart mediation-residual bridge;
6. an operation–trace–closure preorder bridge.

The operation–trace–closure section is included because it extends this same bridge programme. Earlier sections show how triadic schemas induce charted organization and how closure residuals induce learning pressure under faithful readouts. Section 12 adds a different kind of integration: it shows how operations already introduced across the companion theories generate traces, and how traces generate closure-depth preorders. Thus Section 12 is not a new primitive context and not the first cross-closure result in the note. It is another theorem cluster showing how prior constructions jointly determine a new constraint.

0.6 Closure-depth is not linearly classified here

The bridge theorems below should not be read as assigning all charts, traces, residuals, or schema instances to a single linear hierarchy. Some induced structures may be comparable under a closure-depth relation, some may be equivalent, and some may be incomparable. A chart may present a primitive relation carrier, a proper-organization context, a transition family, a path trace, or a residual readout; these presentations need not occupy a simple numerical sequence.

Accordingly, this document does not treat the companion theories as a linear scaffold. Section 12 defines the closure-depth preorder generated by declared admissible operations and traces. That preorder need not be total and need not determine a numerical index. When the notation

$$X \preceq_{\mathrm{cl}} Y$$

is used, it means that Y is obtainable from X by declared closure, charting, transition, trace, compression, or residual-readout operations. It does not mean that X and Y occupy adjacent levels in a fixed stack.

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1. Inherited notation

1.1 Organization structure

From the Theory of Organization, an organization structure consists of a primitive relation carrier together with quotient faces and a zero-boundary resolution map:

$$R \rightsquigarrow A_R, Q_R, Z_R, \rho_R.$$

The zero-boundary face is a bounded ordered structure:

$$(Z_R, \preceq_R, z_R^-, z_R^+),$$

with proper interior

$$Z_R^\circ = \{z \in Z_R : z_R^- \prec_R z \prec_R z_R^+\}.$$

For a threshold $\theta \in Z_R^\circ$, the proper organization layer is

$$\mathrm{Org}_\theta^\circ(R).$$

1.2 Charted organization

From Charted Organization Theory, a chart is a relation-map internal to a primitive carrier. It has carrier

$$C \subsetneq R$$

and face-readout maps

$$\alpha_{\mathcal{C}} : C \to A_{\mathcal{C}}, \qquad \chi_{\mathcal{C}} : C \to Q_{\mathcal{C}}, \qquad \zeta_{\mathcal{C}} : C \to Z_{\mathcal{C}}.$$

When functional, these induce a chart resolution map

$$\rho_{\mathcal{C}} : A_{\mathcal{C}} \times Q_{\mathcal{C}} \to Z_{\mathcal{C}}.$$

For $\theta \in Z_{\mathcal{C}}^\circ$, the chart-relative proper organization layer is

$$\mathrm{Org}_\theta^\circ(\mathcal{C}).$$

An observer atlas is

$$\mathfrak{A}_R = (\mathbf{C}_R, \mathbf{T}_R),$$

where $\mathbf{C}_R$ is a declared class of admissible observer-charts and $\mathbf{T}_R$ is a declared class of admissible chart transitions.

1.3 Recursive chart learning

From Recursive Chart Learning Theory, an available step field at chart $\mathcal{C}$ is

$$\mathrm{Step}_{\mathfrak{A}}(\mathcal{C}),$$

and finite branches are elements of

$$\mathrm{Br}_{\mathfrak{A}}^{<\infty}.$$

For a task τ, an evaluator is a task-mode

$$E_\tau : X_\tau \to W_\tau, \qquad X_\tau \subseteq \mathrm{Br}_{\mathfrak{A}}^{<\infty},$$

with declared admissible region

$$W_\tau^{\mathrm{adm}} \subseteq W_\tau.$$

A learning path trace is an admitted branch.

1.4 Triadic closure

From Triadic Closure and Recursive Residual Theory, the primitive closure schema uses relation-modes

$$\mathsf{R}, \mathsf{O}, \mathsf{Z}.$$

The general boundary–mediation–visibility schema is denoted

$$\Sigma = (B, M, V).$$

A closure residual has three faces:

$$\Omega = (\Omega_B, \Omega_M, \Omega_V),$$

or in the primitive role notation:

$$\Omega_{\mathsf{ROZ}} = (\Omega_{\mathsf{Z}}, \Omega_{\mathsf{O}}, \Omega_{\mathsf{R}}).$$

1.5 Bridge data and bridge adequacy

A **bridge datum** from a source closure context X to a target closure context Y is declared data

$$\mathfrak{B}_{X \to Y}$$

intended to make an inference from X to Y legitimate.

A bridge datum is **adequate** for a theorem $T : X \to Y$ when the hypotheses of T are satisfied by X together with $\mathfrak{B}_{X \to Y}$.

A bridge datum $\mathfrak{B}_2$ is **stronger than** a bridge datum $\mathfrak{B}_1$, written

$$\mathfrak{B}_1 \preceq_{\mathrm{br}} \mathfrak{B}_2,$$

when every preservation condition supplied by $\mathfrak{B}_1$ is also supplied by $\mathfrak{B}_2$. This relation is theorem-relative unless a global bridge preorder is declared.

A bridge datum is **minimal for a theorem** when it is adequate and removing any listed component makes the theorem's conclusion no longer follow from the remaining data.

1.6 Comparison maps, value bridges, and bridge charts

A **comparison map** is declared data used to compare structures that do not already live in the same closure context. Typical examples include a value-face comparison

$$\kappa : Z_1 \to Z_2,$$

an organization transport

$$T : \mathrm{Org}^{\circ}_{\theta}(\mathcal{C}) \to \mathrm{Org}^{\circ}_{\eta}(\mathcal{D}),$$

or a residual readout

$$\Omega : X \to Z_{\Omega}.$$

A comparison map is **bridge data**. It is not an automatic consequence of vocabulary alignment. If a theorem compares values, organizations, losses, residuals, or thresholds drawn from different closure contexts, the comparison map supplying that comparison is part of the bridge declaration.

A comparison map is not automatically a chart. A comparison map becomes **chart-like** only when it is induced by a relation-map with source, target, and value/readout faces of the sort required by Charted Organization Theory. A **bridge chart** is a chart whose declared role is to make two closure contexts comparable. Thus:

$$\text{comparison map} \in \text{bridge data},$$

while

$$\text{bridge chart} = \text{chart-induced comparison data}.$$

Consequently, a value comparison such as $\kappa : Z_1 \to Z_2$ is bridge data but need not by itself supply charted organization. A chart transition or bridge chart supplies more structure because it compares the relevant faces, not only the value spaces.

Proposition 1.7 — Comparison Requires Bridge Declaration

Any cross-closure inference that uses equality, order, compatibility, flatness, loss comparison, or residual comparison across two closure contexts requires a declared comparison map as bridge data.

Proof Without a declared comparison map, the two contexts have no specified common target in which equality, order, compatibility, flatness, loss comparison, or residual comparison can be evaluated. Any such comparison would therefore add undeclared structure to the inference. By the bridge declaration discipline, undeclared cross-context structure cannot license a cross-closure implication. Hence the comparison map is required as bridge data. □

Remark 1.8 — Comparison and charting

A bridge declaration may be lower-dimensional than a chart. It may compare only value faces, only organizations, or only residuals. It becomes a chart-level bridge only when the comparison is realized by a relation-map with enough face structure to induce the relevant charted organization machinery.

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2. Schema-to-chart bridge

Definition 2.1 — Chartability data

Let $\Sigma = (B, M, V)$ be a boundary–mediation–visibility schema instance. A **chartability datum** for Σ is a tuple

$$\mathfrak{C}_\Sigma = (C_\Sigma, A_\Sigma, Q_\Sigma, Z_\Sigma, \alpha_\Sigma, \chi_\Sigma, \zeta_\Sigma, \preceq_\Sigma)$$

such that:

1. C_Σ is a nonempty carrier supplied by the mediation role M;
2. A_Σ and Q_Σ are nonempty face sets supplied by the mediation role or its answer/question decomposition;
3. Z_Σ is a bounded zero-boundary supplied by the boundary role B, with extrema z_Σ^-, z_Σ^+ and nonempty proper interior Z_Σ°;
4. $\alpha_\Sigma : C_\Sigma \to A_\Sigma$, $\chi_\Sigma : C_\Sigma \to Q_\Sigma$, and $\zeta_\Sigma : C_\Sigma \to Z_\Sigma$ are surjective face-readout maps supplied by the visibility role V;
5. the induced resolution is functional: for every $(a, q) \in A_\Sigma \times Q_\Sigma$, the fiber

$$C_{a,q} = \{c \in C_\Sigma : \alpha_\Sigma(c) = a, \chi_\Sigma(c) = q\}$$

is nonempty and $\zeta_\Sigma[C_{a,q}]$ is a singleton.

Definition 2.2 — Induced chart of a schema

Given chartability data $\mathfrak{C}_\Sigma$, define the induced chart

$$\mathcal{C}_\Sigma = (C_\Sigma, A_\Sigma, Q_\Sigma, Z_\Sigma, \alpha_\Sigma, \chi_\Sigma, \zeta_\Sigma, \preceq_\Sigma).$$

Define

$$\rho_\Sigma(a, q) = z$$

iff

$$\zeta_\Sigma[C_{a,q}] = \{z\}.$$

Theorem 2.3 — Schema-to-chart theorem

If a boundary–mediation–visibility schema instance Σ is equipped with chartability data $\mathfrak{C}_\Sigma$, then $\mathcal{C}_\Sigma$ is a chart in the sense of Charted Organization Theory, and $\rho_\Sigma : A_\Sigma \times Q_\Sigma \to Z_\Sigma$ is a well-defined resolution map.

Proof The data of $\mathfrak{C}_\Sigma$ supply a nonempty carrier, nonempty faces, a zero-boundary order, and surjective face-readout maps. The functionality condition states exactly that every answer/question fiber has a unique zero-boundary value. Therefore the construction satisfies the chart relation-map and functionality requirements. The induced map ρ_Σ is well-defined by uniqueness of the singleton $\zeta_\Sigma[C_{a,q}]$. $\square$

Definition 2.4 — Properly chartable schema

A chartable schema instance Σ is **properly chartable** if there exists $\theta \in Z_\Sigma^\circ$ such that

$$\mathrm{Org}_\theta^\circ(\mathcal{C}_\Sigma) \neq \emptyset.$$

Corollary 2.5 — Schema-to-organization theorem

If Σ is properly chartable, then Σ induces a nonempty chart-relative proper organization layer:

$$\mathrm{Org}^{\circ}_{\theta}(\mathcal{C}_{\Sigma}) \neq \emptyset$$

for some $\theta \in Z^{\circ}_{\Sigma}$.

Proof Immediate from Definition 2.4 and the chart construction in Theorem 2.3. $\square$

Remark 2.6 — Bridge character

Theorem 2.3 is a bridge theorem. It does not say every boundary–mediation–visibility schema is chartable. It says that once chartability data are supplied, the charted organization machinery applies.

Definition 2.7 — Schema as chart-class

For a boundary–mediation–visibility schema Σ, define

$$\mathrm{Chart}(\Sigma)$$

as the class of charts $\mathcal{C}$ equipped with chartability data satisfying Definition 2.1 for Σ. Thus Σ is not normally a chart itself. It is a chart-class condition.

Proposition 2.8 — Schema-realization collapse

If a schema instance Σ already includes a carrier, answer face, question face, zero-boundary face, surjective face-readout maps, zero-boundary order, and the functionality condition of Definition 2.1, then Σ determines a canonical chart $\mathcal{C}_{\Sigma}$.

Proof The listed data are exactly the data used in Definition 2.2. The functionality condition gives the induced resolution map. Therefore the schema instance is not merely a classifying pattern; it already contains enough data to be realized as a chart. $\square$

Remark 2.9 — Schema and chart

A schema is a classifying role-pattern. A chart is a relation-map instance satisfying that pattern. In special cases, a schema that already carries all chartability data collapses into a chart. This resolves the ambiguity between schema and chart without identifying every schema with a chart.

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3. Cross-closure instantiations and conditional bridges

Definition 3.1 — Cross-closure instantiation

A **cross-closure instantiation** of the schema-to-chart theorem is a choice of a boundary–mediation–visibility schema instance Σ from one companion theory, together with chartability data $\mathfrak{C}_\Sigma$ sufficient to apply Theorem 2.3.

Example 3.2 — Resolution context

In the Theory of Organization, the resolution context supplies:

$$B = Z_R,$$

$$M = A_R \times Q_R,$$

$$V = \rho_R \text{ or } \Vdash_\theta .$$

The chartability data are:

$$C_\Sigma = R, \qquad A_\Sigma = A_R, \qquad Q_\Sigma = Q_R, \qquad Z_\Sigma = Z_R,$$

with face maps the quotient maps from R to its faces and ζ the zero-boundary read-out. Under the parent theory's total pair coverage and functional resolution axioms, Theorem 2.3 applies.

Example 3.3 — Charted organization context

In Charted Organization Theory, a chart $\mathcal{C}$ already is chartability data:

$$C_\Sigma = C, \qquad A_\Sigma = A_\mathcal{C}, \qquad Q_\Sigma = Q_\mathcal{C}, \qquad Z_\Sigma = Z_\mathcal{C}.$$

The schema-to-chart theorem is therefore identity-level: the charted context is the local chart realization of the schema.

Example 3.4 — Learning branch context

In Recursive Chart Learning Theory, for a declared task τ, a branch domain

$$X_\tau \subseteq \mathrm{Br}_{\mathfrak{A}}^{<\infty}$$

and evaluator

$$E_\tau : X_\tau \to W_\tau$$

supply a chartability instance only after a zero-boundary order is declared on W_τ or on a derived loss/evaluation face. Then:

$$B = W_\tau,$$

$$M = X_\tau,$$

$$V = E_\tau.$$

This instance is not automatic. It is chartable only when W_τ is equipped with proper zero-boundary data and the evaluator is functional over the declared mediation fibers.

Example 3.5 — Closure-residual context

In Triadic Closure and Recursive Residual Theory, a closure residual

$$\Omega = (\Omega_B, \Omega_M, \Omega_V)$$

can be charted if each residual component has a declared zero-boundary value space and the residual readout supplies a functional map from mediation data to residual value. This is a residual-chart bridge, not an automatic consequence of the notation Ω.

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4. Charted inheritance of organization theorems

Theorem 4.1 — Charted inheritance theorem

Let Σ be a properly chartable schema instance. Then every theorem of the Theory of Organization that depends only on the data

$$(A, Q, Z, \rho, \preceq, z^-, z^+)$$

applies to the induced chart

$$\mathcal{C}_\Sigma$$

with

$$(A, Q, Z, \rho) = (A_\Sigma, Q_\Sigma, Z_\Sigma, \rho_\Sigma).$$

In particular, the following apply chartwise:

1. mutual derivation operators;

2. closure/compression operators;
3. proper organization layers;
4. boundary-excision no-gos;
5. persistence support and barcode constructions when the required chain or metric-chain hypotheses are supplied.

Proof Theorems depending only on $(A, Q, Z, \rho, \preceq, z^-, z^+)$ are invariant under replacement of those symbols by any structure satisfying the same axioms. By Theorem 2.3 and proper chartability, $\mathcal{C}_\Sigma$ supplies exactly those data. Therefore the theorems apply by substitution. □

Corollary 4.2 — Charted persistence inheritance

If Σ is properly chartable and Z_Σ satisfies the finite-chain or metric-chain hypotheses required for persistence, then the persistence, barcode, and stability constructions of the Theory of Organization apply to

$$\mathrm{Org}^\circ_\theta(\mathcal{C}_\Sigma).$$

Proof This is the persistence-specific instance of Theorem 4.1. □

Remark 4.3 — Inheritance is not novelty

Theorem 4.1 does not create new organization theorems. It establishes that chartable schema instances inherit existing organization machinery. Its value is cross-closure discipline: it states exactly when vocabulary from Triadic Closure or Recursive Chart Learning can legitimately use the organization machinery.

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5. Residual-to-learning bridge

Definition 5.1 — Residual readout of task loss

Let Ω be a closure residual and let τ be a relation-internal task over a branch domain

$$X_\tau \subseteq \mathrm{Br}^{<\infty}_{\mathfrak{A}}.$$

A **residual readout of task loss** is a map

$$\lambda_\tau : \Omega \times X_\tau \to L_\tau$$

such that

$$\mathrm{Loss}_\tau(p) = \lambda_\tau(\Omega, p)$$

for every $p \in X_\tau$.

Definition 5.2 — Faithful residual-loss readout

A residual readout of task loss is **faithful** if

$$\mathrm{Loss}_\tau(p) = 0$$

implies that branch p preserves every task-visible distinction in D_τ.

Definition 5.3 — Loss-induced evaluator

Given an ordered loss space $(L_\tau, \preceq_L)$, define a loss-induced evaluator

$$E_\tau : X_\tau \to W_\tau$$

by order reversal or a declared monotone transformation of loss. In the simplest case,

$$E_\tau(p) = -\,\mathrm{Loss}_\tau(p)$$

when the codomain supports this notation.

Theorem 5.4 — Residual-to-learning-pressure theorem

Let Ω be a proper closure residual. Suppose:

1. Ω has a faithful residual readout of task loss for task τ;
2. the loss readout induces an evaluator $E_\tau : X_\tau \to W_\tau$;
3. the evaluator has a declared admissible region W_τ^{adm};
4. there exist branches $p, q \in X_\tau$ such that

$$\mathrm{Loss}_\tau(q) \prec_L \mathrm{Loss}_\tau(p).$$

Then the closure residual induces nontrivial learning pressure over X_τ: the evaluator distinguishes at least two branches, and under an admission policy containing both outcomes, the branch comparison can classify one branch as better for task τ than the other.

Proof By hypothesis, $\mathrm{Loss}_\tau(q) \prec_L \mathrm{Loss}_\tau(p)$. Since E_τ is induced monotonically by order reversal or by a declared monotone transformation, the evaluator distinguishes p and q in the corresponding outcome order. If the admissible region contains both outcomes, both branches are admissible, and the evaluator supplies a nontrivial ordering among admitted branches. This is learning pressure in the sense of Recursive Chart Learning Theory. □

Corollary 5.5 — Proper residual alone is insufficient

A proper closure residual does not by itself induce learning pressure. Faithful task-loss readout, branch domain, evaluator, and admission region are required.

Proof Theorem 5.4 uses all four data. Without them, no branch comparison is defined. □

Remark 5.6 — No automatic identity of residual and loss

This theorem does not identify a closure residual with loss. Loss is a possible readout of residual under declared task data. Without a faithful readout, statements about residual cannot be transferred to learning.

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6. Persistence-compatible trace theorem

Definition 6.1 — Organization trace

Let

$$p = \mathcal{C}_0 \xrightarrow{\phi_0} \mathcal{C}_1 \xrightarrow{\phi_1} \cdots \xrightarrow{\phi_{n-1}} \mathcal{C}_n$$

be a learning path trace in an observer atlas. An **organization trace over** p is a sequence

$$O_0, O_1, \ldots, O_n$$

such that

$$O_i \in \mathrm{Org}^{\circ}_{\theta_i}(\mathcal{C}_i)$$

for some threshold schedule

$$\Theta = (\theta_0, \ldots, \theta_n).$$

Definition 6.2 — Persistence-compatible threshold schedule

A threshold schedule $\Theta = (\theta_0, \ldots, \theta_n)$ is **persistence-compatible** with an organization trace $(O_i)_{i=0}^{n}$ if

$$\theta_i \in \mathrm{Supp}(O_i)$$

for every i, where $\mathrm{Supp}(O_i)$ is the persistence support of O_i in chart $\mathcal{C}_i$.

If chart transition ϕ_i includes an order map

$$k_i : Z_{\mathcal{C}_i} \to Z_{\mathcal{C}_{i+1}},$$

then the schedule is **transition-compatible** when

$$\theta_{i+1} = k_i(\theta_i)$$

whenever such threshold transport is declared.

Definition 6.3 — Transported organization trace

An organization trace is **transported** if each transition ϕ_i transports O_i to O_{i+1} under the chart-transition organization transport rule.

Theorem 6.4 — Persistence-compatible trace theorem

Let p be a learning path trace. If $(O_i)_{i=0}^n$ is a transported organization trace over p with persistence-compatible threshold schedule Θ, then every node of the trace remains in the proper organization layer:

$$O_i \in \mathrm{Org}^{\circ}_{\theta_i}(\mathcal{C}_i)$$

for all $i = 0, \dots, n$.

Proof By persistence-compatibility, $\theta_i \in \mathrm{Supp}(O_i)$ for each i. By definition of persistence support, this means O_i is a proper organization at threshold θ_i. Transportedness ensures that the sequence (O_i) is coherent along the chart transitions. Therefore every node remains proper at its declared threshold. □

Corollary 6.5 — Threshold failure blocks proper-trace claims

If for some i,

$$\theta_i \notin \mathrm{Supp}(O_i),$$

then the trace cannot claim that O_i is proper at threshold θ_i.

Proof Immediate from the definition of support. □

Remark 6.6 — Why this is cross-closure

The theorem uses Charted Organization for chart transitions, the Theory of Organization for persistence supports, and Recursive Chart Learning for path traces. It is therefore a genuine cross-closure theorem rather than a restatement internal to one companion theory.

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7. Cross-closure failure theorem

Theorem 7.1 — Cross-closure failure theorem

The following failures block the corresponding cross-closure implication:

1. If a boundary–mediation–visibility schema lacks chartability data, it does not induce a charted organization.
2. If a charted schema is not properly chartable, it does not induce a nonempty proper organization layer.
3. If a closure residual lacks faithful residual-loss readout, it does not induce learning pressure.
4. If a learning path trace lacks persistence-compatible threshold data, it does not preserve proper organization along the trace.

Proof Each item is the contrapositive of a bridge requirement used in Theorem 2.3, Corollary 2.5, Theorem 5.4, or Theorem 6.4. □

Corollary 7.2 — Vocabulary alignment is not enough

The fact that a structure can be verbally assigned boundary, mediation, and visibility roles is not sufficient to transfer results across companion theories. Chartability, faithfulness, and persistence-compatibility data are required.

Proof Immediate from Theorem 7.1. □

Definition 7.3 — Adequate bridge family

For a collection of cross-closure implications $\{X_i \Rightarrow Y_i\}_{i\in I}$, an **adequate bridge family** is a family of adequate bridge data

$$\{\mathfrak{B}_{X_i\to Y_i}\}_{i\in I}.$$

It is **jointly compatible** when the target structure produced by each bridge satisfies the source-side hypotheses of every later bridge with which it is composed.

Theorem 7.4 — Bridge monotonicity

Let $T : X \to Y$ be a cross-closure theorem whose hypotheses are satisfied by bridge data $\mathfrak{B}_1$. If $\mathfrak{B}_1 \preceq_{\mathrm{br}} \mathfrak{B}_2$, then $\mathfrak{B}_2$ also satisfies the preservation hypotheses of T, provided $\mathfrak{B}_2$ does not violate any compatibility condition of T.

Proof By definition, $\mathfrak{B}_2$ supplies every preservation condition supplied by $\mathfrak{B}_1$. Since $\mathfrak{B}_1$ satisfies the theorem hypotheses, the same hypotheses are satisfied by $\mathfrak{B}_2$, except for conditions explicitly requiring compatibility among the supplied data. The stated compatibility clause excludes those failures. □

Theorem 7.5 — Bridge composition

Suppose a source structure X, bridge data $\mathfrak{B}_{X \to Y}$, and theorem T_1 induce a target structure Y. Suppose further that Y, bridge data $\mathfrak{B}_{Y \to Z}$, and theorem T_2 induce Z. If the output of T_1 satisfies the source hypotheses of T_2, then the composite bridge data

$$\mathfrak{B}_{X \to Z} = \mathfrak{B}_{X \to Y} \circ \mathfrak{B}_{Y \to Z}$$

induce Z from X.

Proof The first bridge supplies the hypotheses needed to produce Y. The compatibility condition says that this produced Y is an admissible source for the second theorem. The second bridge then supplies the hypotheses needed to produce Z. Hence the composite bridge induces Z from X. □

Proposition 7.6 — Bridge incompatibility

Two adequate bridges may fail to compose if their target/source data conflict. In particular, if $\mathfrak{B}_{X \to Y}$ produces a threshold schedule, residual readout, or transport structure not accepted by $\mathfrak{B}_{Y \to Z}$, then the composite inference $X \to Z$ is not licensed.

Proof Bridge composition requires the output of the first bridge to satisfy the source hypotheses of the second. If the produced data are not accepted by the second bridge, that hypothesis fails. □

Definition 7.7 — Minimal bridge datum

A bridge datum $\mathfrak{B}_{X \to Y}$ is **minimal** for a theorem $T : X \to Y$ when it is adequate and no proper deletion of one of its required components remains adequate for T.

Remark 7.8 — Conditionality as discipline

The conditionality of the bridge theorems is not a weakness to be hidden. It is the content of the bridge declaration discipline. A bridge theorem states both what can be transported across closure contexts and what data must be supplied before such transport is legitimate.

Definition 7.9 — Source–bridge–target triad

A **source–bridge–target triad** is a triple

$$(X, \mathfrak{B}_{X \to Y}, Y)$$

where X is a source closure context, Y is a target closure context, and $\mathfrak{B}_{X\to Y}$ is the declared bridge data licensing a cross-closure inference from X to Y.

This triad is the cross-closure form of the more intuitive question–learning–answer pattern:

$$\text{question/source} \quad -- \quad \text{learning/bridge} \quad -- \quad \text{answer/target.}$$

The bridge is not an external oracle. It is the declared mediation by which the source becomes readable in the target context.

Theorem 7.10 — Source–Bridge–Target Reduction No-Gos

For a cross-closure inference $(X, \mathfrak{B}_{X\to Y}, Y)$:

1. without X, the bridge has no source context to transport;
2. without Y, the bridge has no target context in which the conclusion can be read;
3. without $\mathfrak{B}_{X\to Y}$, the source and target are only vocabulary-aligned and no legitimate cross-closure inference is licensed.

Thus the source–bridge–target triad is irreducible for bridge-based cross-closure inference.

Proof A bridge datum is defined as data from a source context to a target context. If the source is absent, there is no structure to which the bridge applies. If the target is absent, there is no structure induced by the bridge. If the bridge data are absent, the bridge declaration discipline supplies no comparison, preservation, chartability, readout, or threshold data connecting source and target. Hence no cross-closure inference follows in any of the three reductions. □

Corollary 7.11 — Question–Learning–Answer as Cross-Closure Form

When a question is treated as a source demand, learning as the admitted bridge process, and an answer as a target resolution, the question–learning–answer pattern instantiates the source–bridge–target triad. Its reduction failures are therefore the same: no source demand, no learning bridge, or no target resolution blocks the corresponding inference.

Remark 7.12 — Not a new primitive triad

The source–bridge–target triad does not add a new primitive mechanism. It is the bridge-declaration form of the same boundary–mediation–visibility discipline: the source supplies the unresolved context, the bridge supplies mediation, and the target supplies the context in which the result becomes visible.

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8. Worked internal applications

8.1 Resolution context

The Theory of Organization directly satisfies the schema-to-chart conditions by construction. Therefore the charted organization machinery recovers the parent organization structure when the schema is instantiated by

$$(Z_R, A_R \times Q_R, \rho_R).$$

This is the identity case of the bridge.

8.2 Chart context

For a chart $\mathcal{C}$, the bridge is again direct. The chart already supplies chartability data. The cross-closure content is that any chart satisfying proper organization conditions inherits the parent persistence machinery.

8.3 Learning context

For a task τ, a branch domain and evaluator become chartable only when the evaluator codomain is supplied with zero-boundary order data and the evaluator is treated as a resolution readout. Thus learning data do not automatically form organization data; they become organization data only through a declared evaluation boundary.

8.4 Closure-residual context

A closure residual becomes learning pressure only through a faithful residual-loss readout. This prevents the invalid inference:

$$\text{residual exists} \Rightarrow \text{learning pressure exists.}$$

The valid inference is conditional:

$$\text{residual}+\text{faithful task-loss readout}+\text{branch domain}+\text{admission rule} \Rightarrow \text{learning pressure.}$$

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9. Three-chart comparison and mediation residual

The two-chart transport results of Charted Organization Theory establish when organization can be moved from one chart to another. A cross-closure test of triadic mediation requires one more chart. Three charts allow direct transport to be compared with transport through an intermediate chart.

Definition 9.1 — Three-chart comparison datum

A **three-chart comparison datum** in an observer atlas $\mathfrak{A}_R$ consists of admissible charts

$$\mathcal{C}_0, \mathcal{C}_1, \mathcal{C}_2 \in \mathbf{C}_R$$

and admissible chart transitions

$$\phi_{01} : \mathcal{C}_0 \to \mathcal{C}_1, \qquad \phi_{12} : \mathcal{C}_1 \to \mathcal{C}_2, \qquad \phi_{02} : \mathcal{C}_0 \to \mathcal{C}_2.$$

The transition ϕ_{02} is called the **direct transition**. The composite

$$\phi_{12} \circ \phi_{01}$$

when defined in the declared transition class, is called the **mediated transition**.

Definition 9.2 — Organization transport along a comparison datum

Assume that each transition in a three-chart comparison datum has an induced organization transport operation, written

$$T_{01}, \quad T_{12}, \quad T_{02}.$$

For an organization

$$O_0 \in \mathrm{Org}^{\circ}_{\theta_0}(\mathcal{C}_0),$$

the **direct transport** is

$$T_{02}(O_0),$$

and the **mediated transport** is

$$T_{12}(T_{01}(O_0)).$$

These expressions are defined only when the corresponding transported organizations exist at the relevant thresholds.

Definition 9.3 — Flatness on an organization

A three-chart comparison datum is **flat on** O_0 when both direct and mediated transports are defined and

$$T_{02}(O_0) = T_{12}(T_{01}(O_0)).$$

It is **non-flat on** O_0 when both transports are defined but unequal.

Definition 9.4 — Mediation residual

A **mediation residual readout** for a three-chart comparison datum is a declared map

$$\Omega_{012} : \mathrm{Dom}(\Omega_{012}) \to Z_\Omega$$

where Z_Ω is a zero-boundary residual value space, and where $\mathrm{Dom}(\Omega_{012})$ contains organizations O_0 for which direct and mediated transports are both defined. The residual value space Z_Ω may be declared independently of the visibility-face value spaces of the three charts; if it is intended to interact with those value spaces, the corresponding comparison or transport maps must be declared.

The residual readout is **flat-faithful** when

$$\Omega_{012}(O_0) = z_\Omega^-$$

if and only if the comparison datum is flat on O_0. Values in

$$Z_\Omega^\circ$$

record proper nonzero mediation residuals.

Theorem 9.5 — Three-chart flatness transport theorem

Let a three-chart comparison datum be flat on

$$O_0 \in \mathrm{Org}^\circ_{\theta_0}(\mathcal{C}_0).$$

If the direct and mediated threshold schedules are persistence-compatible, then direct and mediated transport preserve the same proper organization at $\mathcal{C}_2$.

Proof Flatness gives

$$T_{02}(O_0) = T_{12}(T_{01}(O_0)).$$

Persistence-compatibility ensures that both sides are proper organizations at their declared thresholds. Therefore the direct and mediated traces agree as proper organizations in $\mathcal{C}_2$. $\square$

Theorem 9.6 — Non-flat mediation yields path-trace distinction

Let a three-chart comparison datum be non-flat on

$$O_0 \in \mathrm{Org}^{\circ}_{\theta_0}(\mathcal{C}_0).$$

Assume both direct and mediated threshold schedules are persistence-compatible. Then the direct trace

$$\mathcal{C}_0 \xrightarrow{\phi_{02}} \mathcal{C}_2$$

and the mediated trace

$$\mathcal{C}_0 \xrightarrow{\phi_{01}} \mathcal{C}_1 \xrightarrow{\phi_{12}} \mathcal{C}_2$$

have the same source and target charts but do not determine the same transported organization for O_0.

Proof The source and target charts are $\mathcal{C}_0$ and $\mathcal{C}_2$ in both traces. Non-flatness gives

$$T_{02}(O_0) \neq T_{12}(T_{01}(O_0)).$$

Persistence-compatibility ensures both transported outputs are proper. Hence the two traces share endpoints but differ in transported organization. $\square$

Corollary 9.7 — Two charts cannot detect mediation residual

A two-chart comparison containing only

$$\mathcal{C}_0 \xrightarrow{\phi_{02}} \mathcal{C}_2$$

cannot define the mediation residual of Definition 9.4, because no intermediate mediated transport exists to compare with the direct transport.

Proof Definition 9.4 requires both direct and mediated transports. A two-chart comparison has no intermediate chart $\mathcal{C}_1$ and no mediated composite $\phi_{12} \circ \phi_{01}$. □

Corollary 9.8 — Vocabulary-level triadicity is not enough

The presence of three labels $\mathcal{C}_0, \mathcal{C}_1, \mathcal{C}_2$ is not sufficient to define a mediation residual. The direct transition, the mediated transition, organization transport, and a residual readout must be declared.

Proof Immediate from Definitions 9.1–9.4. □

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10. Three-chart no-go catalogue

Theorem 10.1 — No direct transition, no flatness comparison

If a three-chart datum lacks $\phi_{02} : \mathcal{C}_0 \to \mathcal{C}_2$, then direct-versus-mediated flatness is undefined.

Proof Flatness in Definition 9.3 compares $T_{02}(O_0)$ with $T_{12}(T_{01}(O_0))$. Without ϕ_{02}, the transport T_{02} is not defined. □

Theorem 10.2 — No mediated composite, no mediation residual

If a three-chart datum lacks either ϕ_{01}, ϕ_{12}, or their declared composite, then the mediation residual is undefined.

Proof The mediated transport in Definition 9.2 is $T_{12}(T_{01}(O_0))$. This requires both component transports and their composability. □

Theorem 10.3 — No organization transport, no organization-level residual

If transitions exist but do not induce organization transport maps on the relevant proper organization layer, then no organization-level mediation residual can be formed.

Proof Definition 9.4 applies to organizations O_0 for which direct and mediated transports are both defined. If no organization transport is declared, these expressions are undefined. □

Theorem 10.4 — No residual readout, no residual value

If direct and mediated transports differ but no residual readout Ω_{012} is declared, then the comparison determines inequality but not a residual value in any zero-boundary residual space.

Proof Inequality of transports is a statement in the organization layer. A residual value requires a map into Z_Ω. Without such a map, no residual value is defined. □

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11. Integrated three-chart worked example

This example shows schema-to-chart inheritance, three-chart comparison, mediation residual, learning-pressure readout, persistence compatibility, and failure of two-chart detection in one finite setting.

11.1 Charts and organizations

Let there be three admissible charts

$$\mathcal{C}_0, \mathcal{C}_1, \mathcal{C}_2$$

with common zero-boundary

$$Z = [0, 1], \qquad z^- = 0, \qquad z^+ = 1.$$

Each chart has two proper organizations at threshold $\theta = 0.5$:

$$O_0^a, O_0^b \in \mathrm{Org}_{0.5}^{\circ}(\mathcal{C}_0),$$

$$O_1^a, O_1^b \in \mathrm{Org}_{0.5}^{\circ}(\mathcal{C}_1),$$

$$O_2^x, O_2^y, O_2^z \in \mathrm{Org}_{0.5}^{\circ}(\mathcal{C}_2).$$

Assume each listed organization has persistence support

$$(0.2, 0.9].$$

Thus threshold 0.5 is persistence-compatible for all listed organizations.

11.2 Transitions

Declare transitions

$$\phi_{01} : \mathcal{C}_0 \to \mathcal{C}_1,$$

$$\phi_{12} : \mathcal{C}_1 \to \mathcal{C}_2,$$

$$\phi_{02} : \mathcal{C}_0 \to \mathcal{C}_2.$$

Their organization transports are defined by

$$T_{01}(O_0^a) = O_1^a, \qquad T_{01}(O_0^b) = O_1^b,$$

$$T_{12}(O_1^a) = O_2^x, \qquad T_{12}(O_1^b) = O_2^y,$$

and

$$T_{02}(O_0^a) = O_2^x, \qquad T_{02}(O_0^b) = O_2^z.$$

Therefore the datum is flat on O_0^a because

$$T_{02}(O_0^a) = O_2^x = T_{12}(T_{01}(O_0^a)).$$

It is non-flat on O_0^b because

$$T_{02}(O_0^b) = O_2^z \neq O_2^y = T_{12}(T_{01}(O_0^b)).$$

11.3 Mediation residual

Let

$$Z_\Omega = \{0, r, 1\}, \qquad 0 \prec r \prec 1.$$

Define a flat-faithful residual readout by

$$\Omega_{012}(O_0^a) = 0,$$

$$\Omega_{012}(O_0^b) = r.$$

Thus O_0^b has a proper nonzero mediation residual.

11.4 Learning-pressure readout

Let τ be the task that prefers mediated preservation of O_0^b. Define branches

$$p_{\text{med}} = \phi_{12} \circ \phi_{01}, \qquad p_{\text{dir}} = \phi_{02}.$$

Define a loss readout from the residual by

$$\text{Loss}_\tau(p_{\text{med}}) = 0,$$

$$\text{Loss}_\tau(p_{\text{dir}}) = 0.4.$$

Let the evaluator be

$$E_\tau(p) = 1 - \text{Loss}_\tau(p),$$

with admissible region

$$W_\tau^{\text{adm}} = [0.5, 1].$$

Then

$$E_\tau(p_{\text{med}}) = 1,$$

$$E_\tau(p_{\text{dir}}) = 0.6.$$

Both branches are admitted, but the mediated branch is better for task τ. The residual therefore induces learning pressure by Theorem 5.4.

11.5 Calibration and same-endpoint path distinction

Both branches have source $\mathcal{C}_0$ and target $\mathcal{C}_2$, but they transport O_0^b differently. Let calibration loss between these branches be

$$\text{CalLoss}_\tau(p_{\text{med}}, p_{\text{dir}}) = |E_\tau(p_{\text{med}}) - E_\tau(p_{\text{dir}})| = 0.4.$$

At tolerance $\varepsilon = 0.5$, the pair is calibration-stable. At tolerance $\varepsilon = 0.25$, it is calibration-failed.

Thus same source and target charts do not determine the same learning path trace, and calibration depends on the declared tolerance.

11.6 Two-chart insufficiency

If only $\mathcal{C}_0$ and $\mathcal{C}_2$ are retained, the direct transition ϕ_{02} remains, but the mediated path through $\mathcal{C}_1$ is unavailable. The residual Ω_{012} cannot be formed, even though the direct branch is still defined. This illustrates Corollary 9.7.

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12. Operation–trace–closure preorder

This section records a compact formal treatment of closure-depth. It is included to prevent the bridge theorems above from being misread as imposing a simple linear hierarchy. The basic structure is not a numerical index. It is the triad

$$\boxed{\text{operation} \quad — \quad \text{trace} \quad — \quad \text{closure.}}$$

An operation is a local admissible move. A trace is a finite composable record of such moves. Closure is the reachability/stability relation generated by those traces. Depth, when available, is a readout of closure, not primitive data.

Definition 12.1 — Admissible operation system

Let X be a declared class of structures. An **admissible operation system** on X is a class

$$\mathsf{Op} \subseteq \bigsqcup_{X,Y \in \mathsf{X}} \mathrm{Op}(X,Y)$$

whose elements are typed operations

$$u : X \to Y.$$

The typing data $\mathrm{src}(u) = X$ and $\mathrm{tgt}(u) = Y$ are part of the operation data.

The operation system is **composition-closed where declared** when, whenever

$$u : X \to Y, \qquad v : Y \to Z$$

are declared composable operations, there is a declared composite

$$v \circ u : X \to Z.$$

This definition does not require every formally composable pair to have a declared composite. Composition is a declared structural feature, not an automatic background assumption.

Definition 12.2 — Operation trace

Given an operation system Op, an **operation trace** from X_0 to X_n is a finite sequence

$$p = (X_0 \xrightarrow{u_0} X_1 \xrightarrow{u_1} \cdots \xrightarrow{u_{n-1}} X_n)$$

such that each

$$u_i : X_i \to X_{i+1}$$

belongs to Op. The empty trace at X is denoted $\mathrm{id}_X^{\mathrm{tr}}$.
Write

$$\mathrm{Tr}_{\mathsf{Op}}(X, Y)$$

for the class of operation traces from X to Y.

Definition 12.3 — Trace-generated closure

The **trace-generated closure** of X is

$$\mathrm{Cl}_{\mathsf{Op}}(X) = \{Y \in \mathsf{X} : \mathrm{Tr}_{\mathsf{Op}}(X, Y) \neq \emptyset\}.$$

Define the **closure-depth preorder** by

$$X \preceq_{\mathrm{cl}} Y \quad \Longleftrightarrow \quad Y \in \mathrm{Cl}_{\mathsf{Op}}(X).$$

Thus $X \preceq_{\mathrm{cl}} Y$ means that Y is generated from X by at least one admissible operation trace.

Theorem 12.4 — Trace-generated closure is a preorder

Assume empty traces are admitted and the operation system is closed under declared trace concatenation. Then $\preceq_{\mathrm{cl}}$ is reflexive and transitive.

Proof Reflexivity follows because the empty trace $\mathrm{id}_X^{\mathrm{tr}}$ is a trace from X to X. Hence $X \preceq_{\mathrm{cl}} X$.

For transitivity, suppose $X \preceq_{\mathrm{cl}} Y$ and $Y \preceq_{\mathrm{cl}} Z$. Then there is a trace $p : X \to Y$ and a trace $q : Y \to Z$. By declared trace concatenation, $q * p : X \to Z$ is a trace. Therefore $X \preceq_{\mathrm{cl}} Z$. $\square$

Definition 12.5 — Closure-equivalence

Two structures are **closure-equivalent** when

$$X \sim_{\text{cl}} Y \quad \Longleftrightarrow \quad X \preceq_{\text{cl}} Y \text{ and } Y \preceq_{\text{cl}} X.$$

Closure-equivalence means that each structure is reachable from the other by admissible traces. It does not require literal identity.

Definition 12.6 — Depth readout

A **depth readout** is a map

$$d_{\text{cl}} : \mathsf{X} \to P$$

into a preorder $(P, \leq_P)$ such that

$$X \preceq_{\text{cl}} Y \quad \Longrightarrow \quad d_{\text{cl}}(X) \leq_P d_{\text{cl}}(Y)$$

whenever the intended reading is monotone depth. A numerical depth, if used, is a special case of this readout. It is not part of the closure preorder itself.

Theorem 12.7 — No operation, no nontrivial trace

If Op contains only empty identity traces, then

$$\text{Cl}_{\mathsf{Op}}(X) = \{X\}$$

for every X. Hence the closure-depth preorder is equality up to identity traces.

Proof If no non-identity operation exists, then the only trace starting at X is the empty trace ending at X. Therefore the only element of $\text{Cl}_{\mathsf{Op}}(X)$ is X. □

Theorem 12.8 — No trace, no path-sensitive closure

A set of operations without trace data cannot distinguish two derivations with the same source and target.

Proof Path-sensitive closure requires the data of which finite sequence of operations was used. If only source and target reachability are retained, two distinct traces

$$p, q \in \text{Tr}_{\mathsf{Op}}(X, Y)$$

collapse to the same relation $X \preceq_{\text{cl}} Y$. Therefore trace-sensitive distinctions are unavailable without trace data. □

Theorem 12.9 — No closure, no depth

A trace family without a declared closure relation does not determine a depth preorder.

Proof Depth compares structures by generated reachability or rank. Such comparison requires a relation such as $\preceq_{\text{cl}}$. A raw class of traces may list histories, but until one declares which targets count as generated, reachable, stable, or closed under those traces, no preorder on structures is determined. □

Theorem 12.10 — No canonical numerical depth without rank data

A closure-depth preorder need not be a total order. Therefore a numerical depth index is not determined by closure alone.

Proof Let $\mathsf{X} = \{A, B, C\}$. Let Op contain an operation $A \to B$, an operation $A \to C$, and identity traces, but no operation or trace from B to C or from C to B. Then

$$A \preceq_{\text{cl}} B, \qquad A \preceq_{\text{cl}} C,$$

but neither

$$B \preceq_{\text{cl}} C$$

nor

$$C \preceq_{\text{cl}} B$$

holds. Hence the preorder is not total. A numerical depth index would require extra ranking or tie-breaking data. □

For a programme-internal reading, take B and C to be two charts both induced from a shared schema instance, but with no admissible chart transition between them. Both are closure-reachable from the schema instance, while neither is closure-reachable from the other. The same closure-depth incomparability occurs without assigning them to separate numerical layers.

Corollary 12.11 — Depth is a readout of closure, not primitive structure

Any numerical depth assignment depends on a declared rank or evaluator on $\preceq_{\text{cl}}$. Without such a readout, closure provides a preorder, not a linear stratification.

Theorem 12.12 — Operation–trace–closure reduction no-gos

The triad

$$(\text{operation}, \text{trace}, \text{closure})$$

is reduction-sensitive in the following sense:

1. without operations, no nontrivial traces are generated;
2. without traces, path-sensitive closure is unavailable;
3. without closure, traces do not induce depth, reachability, or generated stability.

Proof Clause 1 is Theorem 12.7. Clause 2 is Theorem 12.8. Clause 3 is Theorem 12.9. □

Remark 12.13 — Relation to the boundary–mediation–visibility schema

The operation–trace–closure pattern is another instance of the boundary–mediation–visibility schema, but it should not be treated as a new primitive context. Operations supply local mediation. Traces supply visible path-accounting. Closure supplies the boundary/stability relation generated by those traces. Closure-depth is therefore a derivative readout of operation traces, not an additional primitive ontology.

Remark 12.14 — Relation to this note

The bridge theorems in Sections 2–11 use declared companion theories only where they are needed for specific cross-closure implications. The closure-depth preorder defined here explains why the programme should not be understood as a simple linear scaffold. Some induced structures may be comparable, some closure-equivalent, and some incomparable under $\preceq_{\mathrm{cl}}$.

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13. Adjacent frameworks and import boundaries

This note uses bridge, transport, closure, and residual language that has close neighbors in existing mathematics. These neighbors are not premises of the note, but they locate the results.

13.1 Functoriality and transport

A bridge theorem of the form

$$X + \mathfrak{B}_{X \to Y} \Rightarrow Y$$

resembles a structure-preserving functor only when the source and target contexts have been organized into categories and the bridge data preserve identities and composition. This note does not assume such category structure globally. The charted inheritance theorem is functorial in spirit because it transports theorems that depend only on organization-structure data, but no general functorial semantics is claimed.

13.2 Adjunction-like structure

The schema-to-chart bridge and the chart-as-schema reading form an adjunction-like pair of directions: chartability realizes a schema as a chart, while a chart can be read as an instantiation of a schema. This note does not prove an adjunction. To do so one would need explicit categories of schemas and charts, together with unit and counit maps satisfying triangle identities.

13.3 Closure operators and preorder generation

The operation–trace–closure construction is a standard kind of reachability closure. The contribution here is not the invention of preorder generation by traces. It is the use of that closure to prevent a linear layer reading of the companion theories and to identify the bridge data needed for closure-depth claims.

13.4 Persistence and dynamic compatibility

The persistence-compatible trace theorem links two structures usually treated separately: persistence supports from a filtration and dynamic traces through charts. The theorem states that a trace remains proper only while its threshold schedule stays inside the transported persistence supports. This can be read as a dynamic compatibility condition for persistence data, not as a new persistence-stability theorem.

13.5 Cocycle and cohomological neighbors

The three-chart mediation residual has the same formal location as cocycle or holonomy data: it measures failure of direct and mediated transport around a triangle to agree. This note does not build a cohomology theory. It only states the bridge conditions under which a residual readout is meaningful.

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14. Claims and non-claims

14.1 Claims

This document claims:

1. cross-closure inference requires declared bridge data;
2. bridge data can be compared, composed, and fail to compose under explicit compatibility conditions;
3. a boundary–mediation–visibility schema induces charted organization only under chartability data;
4. properly chartable schema instances inherit organization machinery chartwise;
5. closure residuals induce learning pressure only under faithful task-loss readout;

6. learning traces preserve proper organization only under persistence-compatible threshold schedules;
7. comparison maps are bridge data; chart-like comparison requires chart-induced face structure;
8. source–bridge–target is the bridge-declaration form of question–learning–answer;
9. three-chart mediation residuals are defined only when direct and mediated transports can be compared;
10. closure-depth is generated by operation traces and is generally a preorder, not a canonical numerical index;
11. vocabulary alignment across closure contexts is not enough for cross-closure implication.

14.2 Non-claims

This document does not claim:

1. every triadic schema is chartable;
2. every residual is a loss;
3. every path trace preserves proper organization;
4. every task-loss is faithful;
5. every external triadic framework instantiates this programme;
6. every comparison map is automatically a chart;
7. closure-depth is always a total order or numerical index;
8. any physical theory has been derived.

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15. Summary

The preceding theory documents established organization, charted visibility, recursive learning traces, and triadic closure residuals. This document supplies explicit cross-closure bridge theorems, including a three-chart comparison theorem where direct and mediated chart transport can be compared.

The central message is:

$$\boxed{\text{cross-closure implication requires declared bridge data.}}$$

A triadic closure schema can become a charted organization, but only when chartability data are supplied. A closure residual can become learning pressure, but only when faithfully read as task loss. A learning trace can preserve proper organization, but only when its threshold schedule remains inside persistence supports. A three-chart comparison can expose mediation residual, but only when direct and mediated transports are both declared and comparable. A comparison map is itself bridge data, and it becomes chart-level only when induced by a relation-map with the relevant face structure. A closure-depth claim can be made only when admissible operations and traces generate a declared closure preorder.

Thus the programme is not merely a collection of structurally similar companion theories. Under declared bridge conditions, structures generated in one closure context can constrain structures governed by another, and three-chart comparisons can detect residual structure invisible to two-chart transport.

AI-assistance disclosure

This document was developed with AI-assisted drafting and review support. The author directed the research programme, selected and revised the mathematical and methodological content, and is responsible for all claims and errors.